

Convex optimisation — A primer

ECMM422

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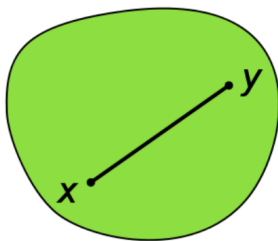
Winter 2026



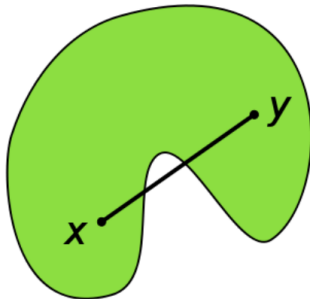
Convex sets

A set C is convex if the line between any two points in C lies in C , i.e. if for any $x, y \in C$ and any $0 \leq t \leq 1$ we have:

$$tx + (1 - t)y$$



Convex set



Non-convex set

Convex sets: examples

Empty sets, singletons and $\mathbb{R}^n$

The empty set $\emptyset$, any single point (i.e. singleton) $\{x_0\}$ and the whole space $\mathbb{R}^n$ are convex

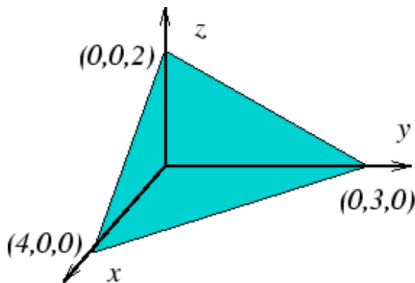
Lines

Any line or line segment is convex

Planes

(Hyper) Planes

$\{x \mid a^T x = b\}$ are convex.



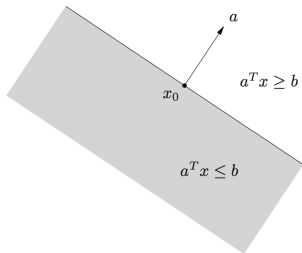
Convex sets: examples

Euclidean ball

$$B(x_c, r) = \{x \mid \|x - x_c\| \leq r\}$$

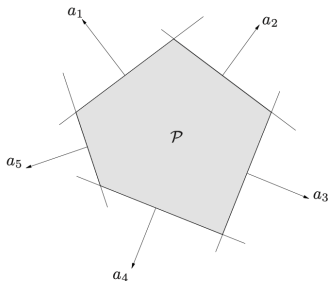
Halfspaces

Hyperplanes divide the entire space into two halfspaces of the form $\{x \mid a^T x \leq b\}$. Halfspaces are convex.

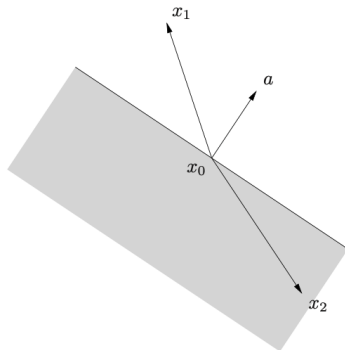


Polyhedra

Polyhedra are the intersection of halfspaces and are therefore convex



Halfspaces

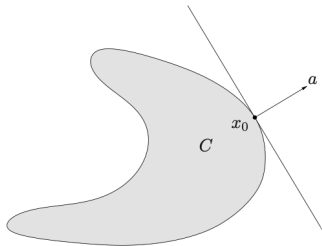


The halfspace can also be expressed as

$$\{x | a^T(x - x_0) \leq 0\}$$

with the geometric interpretation: the halfspace consists of x_0 plus any vector that makes an obtuse (or right) angle with the (outward normal) vector a

Supporting Hyperplane

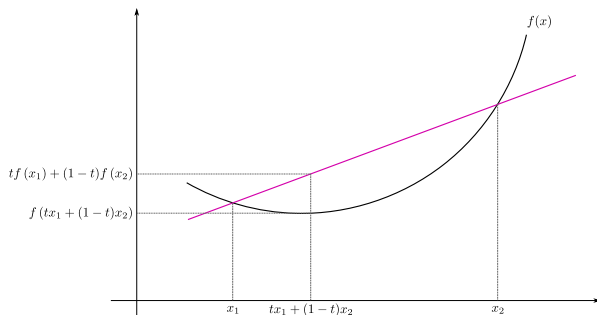


A **supporting hyperplane** for a generic set C at point x_0 is an hyperplane that defines a halfspace that contains C and that is tangent to C in x_0

$$\{x | a^T(x - x_0) = 0\}$$

Supporting hyperplane theorem: for any **convex** set C and **any** point x_0 on its border there exist a supporting hyperplane to C at x_0

Convex functions

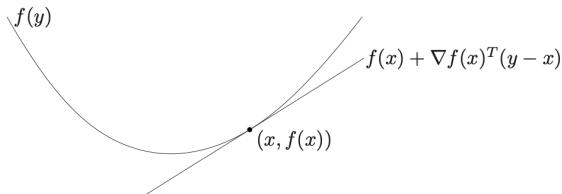


wikipedia.org

A function $f(x)$ is **convex** on $\mathcal{X}$ iff it satisfies **Jensen's inequality** saying that the secant line lies above the graph of $f(x)$. For any $x_1, x_2 \in \mathcal{X}$:

$$tf(x_1) + (1-t)f(x_2) \geq f(tx_1 + (1-t)x_2) \quad \text{for all } 0 \leq t \leq 1$$

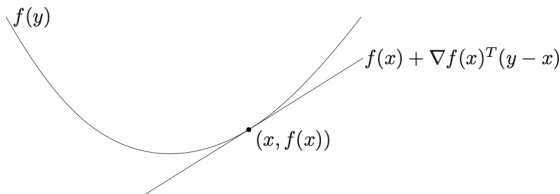
First order conditions



If f is differentiable (i.e. its gradient $\nabla f(x)$ exist at each point) then an equivalent condition is:

$$f(y) \geq f(x) + \nabla f(x)^T(y - x)$$

First order conditions



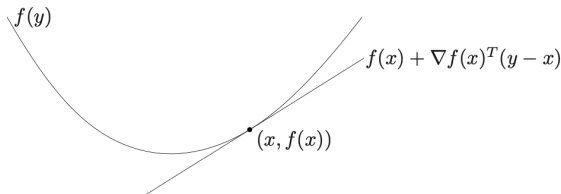
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I.e. for a convex function, the first-order Taylor approximation is a global underestimator of the function

So, for x^* such that $\nabla f(x^*) = 0$, then $f(y) \geq f(x^*)$, i.e. x^* is a global minimum for f

First order conditions



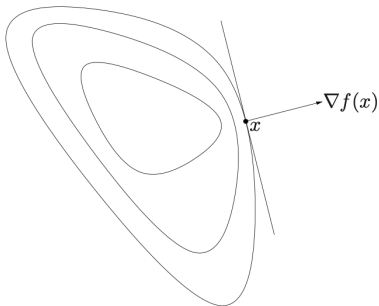
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Main property: convex functions have only one global minimum



The condition

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

has the following geometric interpretation:

$\nabla f(x)$ defines a supporting hyperplane for the sublevel set $\{y | f(y) \leq f(x)\}$

Second order conditions

If f is twice differentiable (i.e. its hessian $\nabla^2 f(x)$ exist at each point) then an equivalent condition is:

$$\nabla^2 f(x) \succeq 0$$

I.e. the Hessian is positive semidefinite

Geometrically it means that the graph of the function have positive (upward) curvature

Optimization problems

In an optimization problem we want to find an x^* that minimizes an objective function among all x that satisfy certain conditions.

An optimisation problem has the following form:

$$\begin{aligned} \min \quad & f_0(x) \\ \text{such that} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_j(x) = 0, \quad j = 1, \dots, p \end{aligned}$$

where:

- ▶ f_0 is the objective function
- ▶ f_i are the inequality constraints functions
- ▶ h_j are the equality constraints functions

If there are no constraints $m = p = 0$ the problem is **unconstrained**

A point x is **feasible** if it satisfies all constraints

Convex optimization problems

A convex optimization problem has the form

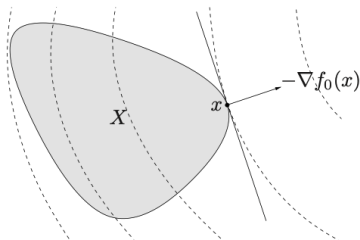
$$\begin{aligned} \min \quad & f_0(x) \\ \text{such that} \quad & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & a_j^T(x) = b_j, \quad j = 1, \dots, p \end{aligned}$$

where the objective and the constraints are all convex:

- ▶ f_0 and f_i are convex functions
- ▶ the equality constraints are affine

Note that the feasibility set is convex

Optimality criterion



The condition

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

has the following geometric interpretation:

$-\nabla f_0(x)$ defines a supporting hyperplane to the feasible set at x

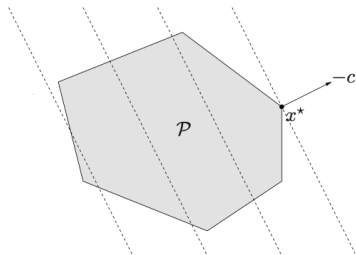
Linear optimization problems

A linear optimization problem has the form

$$\begin{aligned} \min \quad & c^T x + d \\ \text{such that} \quad & Gx \preceq h \\ & Ax = b, \end{aligned}$$

where the objective and the constraints are all affine:

- ▶ $G \in \mathbb{R}^{m \times n}$
- ▶ $A \in \mathbb{R}^{p \times n}$



Quadratic optimization problems

A quadratic optimization problem has the form

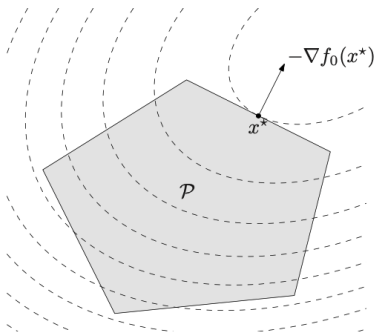
$$\min \frac{1}{2}x^T Px + q^T x + r$$

$$\text{such that } Gx \preceq h$$

$$Ax = b,$$

where:

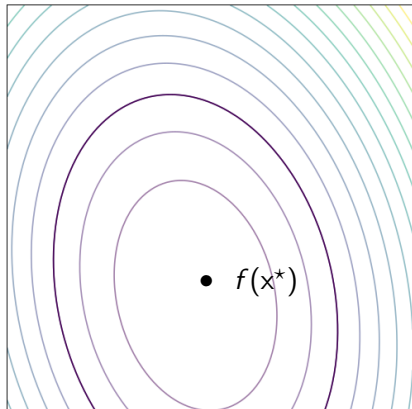
- ▶ $P \in S_+^n$
- ▶ $G \in \mathbb{R}^{m \times n}$
- ▶ $A \in \mathbb{R}^{p \times n}$



Unconstrained optimisation

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex.
Unconstrained minimisation
problem:

$$\min_{x \in \mathbb{R}^d} f(x)$$



Then x^* is the minimiser of f iff

$$\nabla f(x^*) = \left(\frac{\partial f(x^*)}{\partial x_1}, \frac{\partial f(x^*)}{\partial x_2}, \dots, \frac{\partial f(x^*)}{\partial x_d} \right) = 0$$

Minimisation with equality constraints

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and let $h_i(x) = 0$, $i = 1, \dots, p$ be equality constraints. Then the minimiser of the constrained minimisation problem is given by the minimiser of the *Lagrangian*

$$L(x, \nu) = f(x) + \sum_{i=1}^p \nu_i h_i(x)$$

where ν_i are referred to as *Lagrange multipliers* (dual variables)

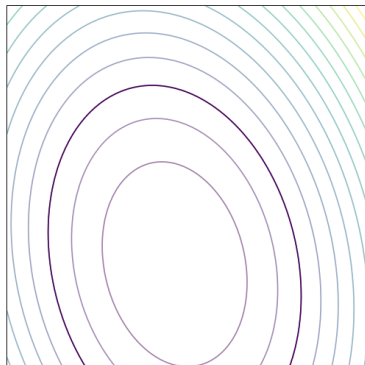
The *Lagrangian dual* function is the minimisation of the Lagrangian over x :

$$g(\nu) = \min_x \left[f(x) + \sum_{i=1}^p \nu_i h_i(x) \right]$$

Strong duality: the **maximum** of $g(\nu)$ is equal to the constrained **minimum** of $f(x)$

Geometric interpretation

- ▶ $\nabla f(x)$ is perpendicular to contours
- ▶ $\nabla h(x)$ is perpendicular to $h(x) = 0$
- ▶ The contour of $f(x)$ that just touches $h(x) = 0$ corresponds to the constrained minimum
- ▶ Therefore $\nabla f(x)$ and $\nabla h(x)$ are parallel at x^*
- ▶ Therefore $\nabla f(x^*) = \nu \nabla h(x^*)$
- ▶ Therefore

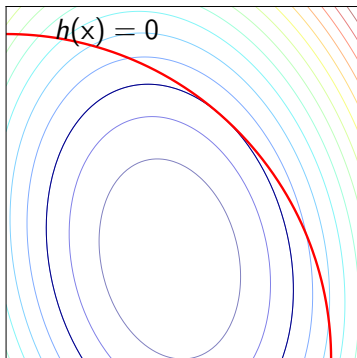


$$\nabla L(x, \nu) = \nabla f(x) + \nu \nabla h(x) = 0$$

finds the constrained minimum

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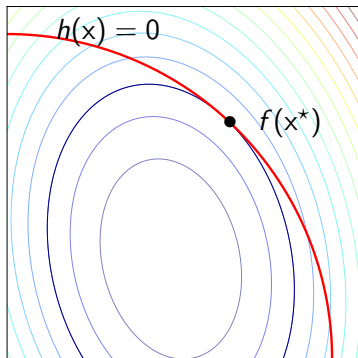


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$$\nabla L(x, \nu) = \nabla f(x) + \nu \nabla h(x) = 0$$

finds the constrained minimum

Langrangian duality

Define the *Langrangian*

$$L(\mathbf{x}, \boldsymbol{\nu}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x})$$

where λ_j and ν_j are referred to as *Langrange multipliers* (dual variables)

The *Langrangian dual* function is the minimisation of the Langrangian over $\mathbf{x}$:

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \min_{\mathbf{x}} \left[f(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{i=1}^p \nu_i h_i(\mathbf{x}) \right]$$

The Lagrangian *dual problem* is defined by

$$\begin{aligned} & \max_{\lambda, \nu} g(\lambda, \nu) \\ & \text{such that } \lambda_i \geq 0, \quad i = 1, \dots, m \end{aligned}$$

Strong duality

Let x^* be any primal optimal and (λ^*, ν^*) be the dual optimal. Under certain conditions on the constraints then the maximum of the dual problem is exactly the minimum of the primal function:

$$g(\lambda^*, \nu^*) = f_0(x^*)$$

Lagrangian Duality: KKT conditions

The Karush-Kuhn-Tucker conditions generalise the Lagrangian multipliers to inequality constraints.

If the f_i are **convex** and the h_i are **affine**, then $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is an optimal solution iff

$$f_i(\mathbf{x}_i^*) \leq 0, \quad i = 1, \dots, m$$

$$h_j(\mathbf{x}_j^*) = 0, \quad j = 1, \dots, p$$

$$\lambda_i^* \geq 0, \quad i = 1, \dots, m$$

$$\lambda_i^* f_i(\mathbf{x}_i^*) = 0, \quad i = 1, \dots, m$$

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla f_i(\mathbf{x}^*) + \sum_{i=1}^p \nu_i \nabla h_i(\mathbf{x}^*) = 0$$

To solve the primal minimisation problem

- ▶ Write down the dual problem (max the Lagrangian dual)
- ▶ Solve the dual using the KKT conditions

Example

Primal problem

$$\min \frac{x^2}{2} \text{ subject to } 2x + 5 \geq 0$$

Example

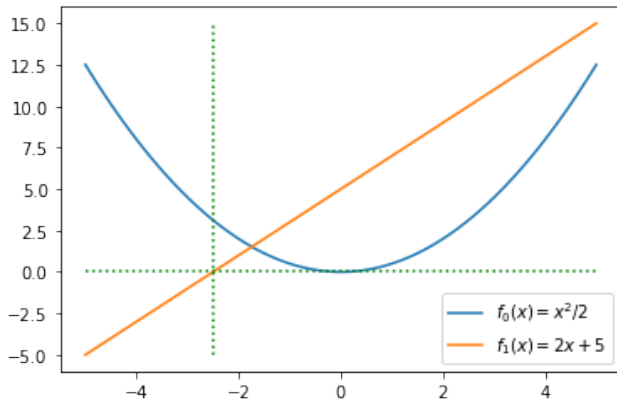
Primal problem

$$\min \frac{x^2}{2} \text{ subject to } 2x + 5 \geq 0$$

Identify with general problem formulation:

- ▶ $f_0(x) = \frac{x^2}{2}$ is convex
- ▶ $f_1(x) = -2x - 5$

Example



Example

- 1 Write down the Lagrangian

$$L(x, \lambda) = f_0(x) + \lambda f_1(x) = \frac{x^2}{2} + \lambda(-2x - 5)$$

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$$L(x, \lambda) = f_0(x) + \lambda f_1(x) = \frac{x^2}{2} + \lambda(-2x - 5)$$

- 2 Lagrangian dual

$$g(\lambda) = \min_x L(x, \lambda)$$

Example

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- 2 Lagrangian dual

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Example

- 1 Write down the Lagrangian

$$L(x, \lambda) = f_0(x) + \lambda f_1(x) = \frac{x^2}{2} + \lambda(-2x - 5)$$

- 2 Lagrangian dual

$$\begin{aligned} g(\lambda) &= \min_x \frac{x^2}{2} + \lambda(-2x - 5) \\ \frac{\partial L(x, \lambda)}{\partial x} &= 0 \\ x^* - 2\lambda &= 0 \\ x^* &= 2\lambda \end{aligned}$$

Example

- 1 Write down the Lagrangian

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- 2 Lagrangian dual

$$g(\lambda) = \min_x \frac{x^2}{2} + \lambda(-2x - 5) = -2\lambda^2 - 5\lambda$$

Example

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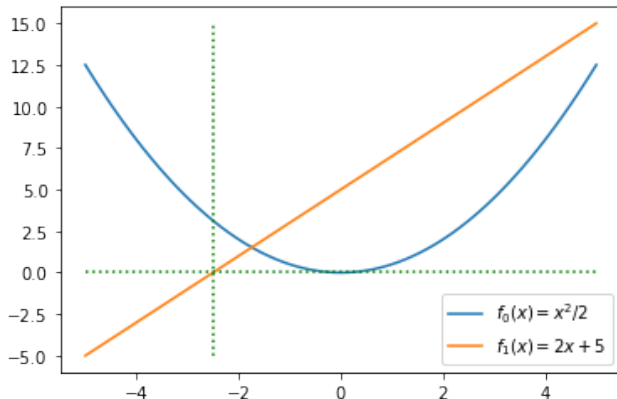
- 2 Lagrangian dual

$$g(\lambda) = \min_x \frac{x^2}{2} + \lambda(-2x - 5) = -2\lambda^2 - 5\lambda$$

- 3 Lagrangian dual problem

$$\begin{aligned} \max_{\lambda} \quad & g(\lambda) \quad \text{such that } \lambda \geq 0 \\ \implies \quad & \lambda^* = 0 \end{aligned}$$

Example



At $\lambda^* = 0$

$$g(\lambda^*) = f_0(x^*) = 0$$

From the Lagrangian dual: $x^* = 2\lambda^* = 0$

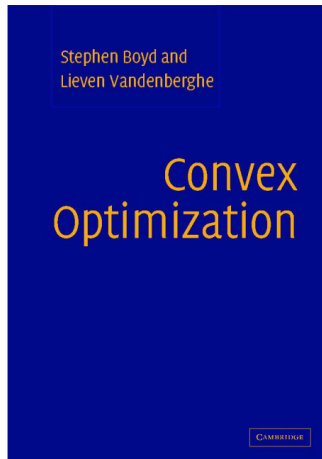
Summary

To solve the primal minimisation problem

- ▶ Write down the dual problem (maximise the Lagrangian dual)
- ▶ Solve the dual using the KKT conditions

Duality gap

- ▶ With weaker conditions on f_0 and constraints there may be a *duality gap*: $g(\lambda^*, \nu^*) \geq f_0(x^*)$
- ▶ In this case the Lagrangian dual gives only a bound



Boyd and Vandenberghe (2004). *Convex Optimization*. Cambridge University Press. http://www.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf